Hints: Inequalities – 1. The Cauchy–Bunyakovsky inequality as a technique

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Discrete form: $P(\lambda) = \sum_i (a_i - \lambda b_i)^2 \geqslant 0$ for all real λ is a quadratic in λ (if not all $b_i = 0$), so its discriminant $4\big(\sum a_i b_i\big)^2 - 4\sum a_i^2 \sum b_i^2$ is $\leqslant 0$. Equality means P has a real root λ, i.e. $a_i = \lambda b_i$ for all i.

Hint 2. The integral form and the inner product form are word for word the same with $P(\lambda) = \int (f - \lambda g)^2$ and $P(\lambda) = \langle f - \lambda g, f - \lambda g\rangle$.

Hint 3. Engel form: apply the discrete inequality to the vectors $\big(a_i/\sqrt{b_i}\big)$ and $\big(\sqrt{b_i}\big)$: $\big(\sum a_i\big)^2 = \big(\sum \frac{a_i}{\sqrt{b_i}}\sqrt{b_i}\big)^2 \leqslant \sum \frac{a_i^2}{b_i} \sum b_i$.

Problem 2

Hint 1. The two conditions give $\int_0^1 (ax + b) f(x)\,dx = a + b$ for all real a, b. Apply Cauchy–Bunyakovsky to f and a linear function $ax + b$.

Hint 2. Normalise $a+b = 2$: then $4 \leqslant \int_0^1 (ax+b)^2\,dx \cdot \int_0^1 f^2$. Choose a, b so that $\int_0^1 (ax+b)^2\,dx = \frac{(a+b)^3 - b^3}{3a} = 1$; together with $a + b = 2$ this gives $b^3 - 3b - 2 = (b + 1)^2(b - 2) = 0$.

Hint 3. The root $b = 2$, $a = 0$ is useless (the equality case $f = \text{const}$ contradicts the conditions); take $b = -1$, $a = 3$. Then $\int f^2 \geqslant 4$, with equality for $f(x) = 6x - 2$.

Problem 3

Hint 1. As before, $\int_0^1 p(x) f(x)\,dx = \sum_k a_k$ for every polynomial $p = \sum_{k<n} a_k x^k$. So $\big(\sum a_k\big)^2 \leqslant \int_0^1 p^2 \cdot \int_0^1 f^2$, where $\int_0^1 p^2 = a^T H_n a$ with the Hilbert matrix $H_n = \big(\frac{1}{i+j+1}\big)_{i,j=0}^{n-1}$.

Hint 2. The best choice of p is the one for which equality is possible, i.e. f proportional to p; the conditions on f then say $H_n a$ is proportional to $(1, \dots, 1)^T$. Take $a = H_n^{-1}\mathbf{1}$: then $\sum a_k = \mathbf{1}^T H_n^{-1}\mathbf{1} = a^T H_n a =: s$ and the inequality gives $\int f^2 \geqslant s^2/s = s$.

Hint 3. It remains to show that the sum of all entries of H_n^{-1} is n^2. One way: $\int_0^1 pq = q(1)$ for all q of degree $< n$, so p is the reproducing kernel at 1 for the orthonormal shifted Legendre polynomials $\sqrt{2k+1}P_k(2x-1)$, and $\int p^2 = \sum_{k<n}(2k+1) = n^2$. For $n = 2$ the conditions are $\int f = \int xf = 1$ and you recover Problem 2.

Problem 4

Hint 1. Substitute $a_i = \frac{1}{1+x_i}$: then $\sum a_i = 1$ and $x_i = \frac{1-a_i}{a_i} = \frac{\sum_{j\neq i} a_j}{a_i}$.

Hint 2. Lower bound the left side: $\sqrt{\sum_{j\neq i} a_j} \geqslant \frac{1}{\sqrt{n-1}} \sum_{j\neq i} \sqrt{a_j}$, so

$$\sum_i \sqrt{x_i} \geqslant \frac{1}{\sqrt{n-1}} \sum_j \sqrt{a_j} \sum_{i\neq j} \frac{1}{\sqrt{a_i}}$$

(swap the order of summation).

Hint 3. By the Engel form, $\sum_{i\neq j}\frac{1}{\sqrt{a_i}} \geqslant \frac{(n-1)^2}{\sum_{i\neq j}\sqrt{a_i}}$, and by Cauchy–Bunyakovsky again $\sum_{i\neq j}\sqrt{a_i} \leqslant \sqrt{(n-1)\sum_{i\neq j}a_i}$. Put these together: $\frac{\sqrt{a_j}}{\sqrt{\sum_{i\neq j}a_i}} = \frac{1}{\sqrt{x_j}}$, and the constants combine to $(n-1)$.

Problem 5

Hint 1. Use $abc = 1$ to rewrite each term as a square over something: $\frac{1}{a^3(b+c)} = \frac{(bc)^2}{a(b+c)}$.

Hint 2. Engel form: $\sum\frac{(bc)^2}{ab+ac} \geqslant \frac{(ab+bc+ca)^2}{2(ab+bc+ca)} = \frac{ab+bc+ca}{2}$. It remains to prove $x+y+z \geqslant 3$ for $x = ab$, $y = bc$, $z = ca$, $xyz = 1$, without AM-GM.

Hint 3. Since $xyz = 1$, $xy+yz+zx = \frac{1}{x}+\frac{1}{y}+\frac{1}{z}$, so Cauchy–Bunyakovsky gives $(x+y+z)(xy+yz+zx) \geqslant 9$. Also $(x+y+z)^2 \geqslant 3(xy+yz+zx)$ (from $x^2+y^2+z^2 \geqslant xy+yz+zx$, again Cauchy–Bunyakovsky). Multiply: $(x+y+z)^3 \geqslant 27$.